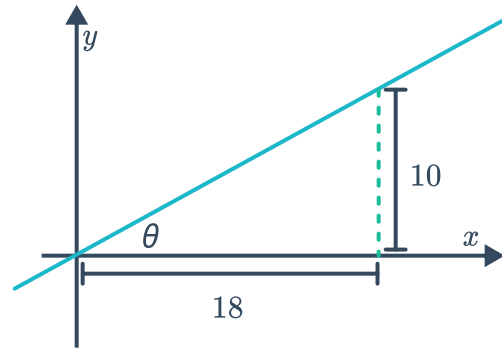


## Angles of inclination

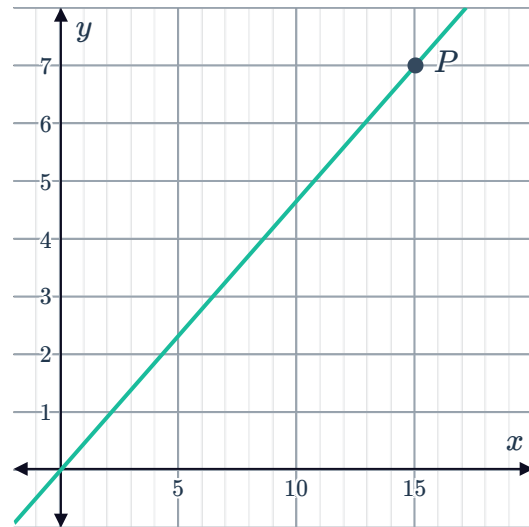


- 1 Find the slope of a line that is inclined at an angle of  $50^\circ$  to the positive  $x$ -axis. Round your answer to one decimal place.
- 2 Find the measure of the angle of inclination of a line to the positive  $x$ -axis, whose slope is 4. Round your answer to the nearest degree.
- 3 Consider the given line with an angle of inclination of  $\theta$  with the positive  $x$ -axis:

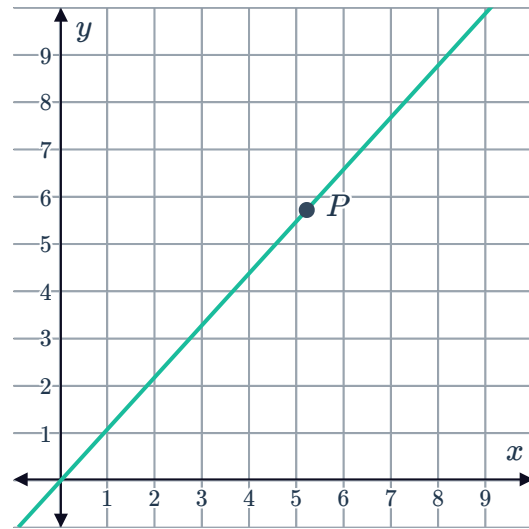
- a Find the slope of the line.
- b Now, find the measure of  $\theta$ . Round your answer to two decimal places.



- 4 In the following graph, point  $P(15, 7)$  lies on the line passing through the origin. Find the measure of  $\theta$ , the angle of inclination of the line with the positive  $x$ -axis. Round your answer to two decimal places.

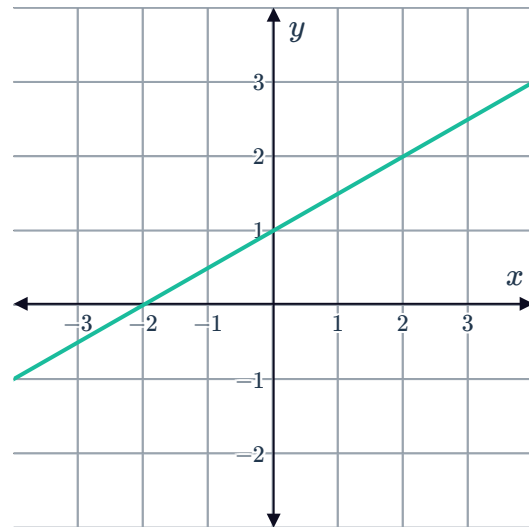


- 5 In the following graph, point  $P(5.2, 5.72)$  lies on the line passing through the origin. Find the measure of  $\theta$ , the angle of inclination of the line with the positive  $x$ -axis. Round your answer to two decimal places.

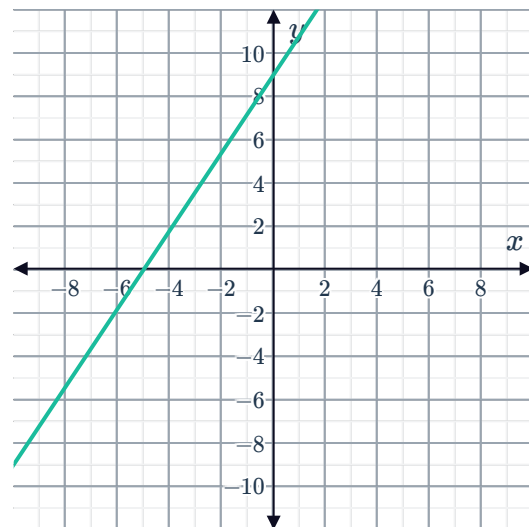


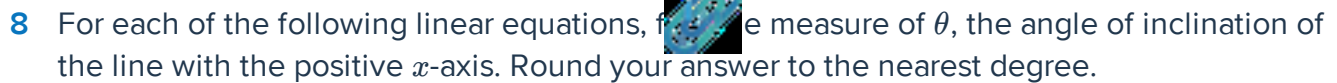
- 6 Consider the graph of the line  $y = \frac{1}{2}x + 1$  given and the triangle whose vertices are the origin, the  $x$ -intercept and the  $y$ -intercept.  $\theta$  is the acute angle in the triangle that the line makes with the  $x$ -axis.

- a State the slope of the line.
- b Find the value of  $\tan \theta$ .



- 7 Consider the following graph: Find the measure of  $\theta$ , the angle of inclination of the line with the positive  $x$ -axis, if the  $x$ -intercept is  $-5$  and the  $y$ -intercept is  $9$ . Round your answer to two decimal places.





**b**  $5x + 6y - 2 = 0$

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